2. The Manifold Memory Architecture

Here is the blueprint for rewriting your memory system (replacing the naive skills.py).

Brain Analogue	System Component	Mathematical Function	Role in AI
Hippocampus	Episodic Buffer	$E = \{x_1, x_2, \dots\}$	Temporarily stores the full 2,000-word derivation.
Consolidation	The Compressor	Quotient map $\pi: E \to S$	Strips context, extracts the topological invariant (the axiom).
Neocortex	Semantic Manifold	Riemannian Manifold $\mathcal{M}$	Stores the compressed axioms using Information Geometry.
Prefrontal Cortex	Attention / Retrieval	Tangent space projection	Injects only the relevant, dense axiom into the Master Agent.

Step A: The Episodic Buffer (Ingestion)

When the Master Agent solves (or fails) a problem, the entire raw trace (prompt, code, output) is dumped into a temporary buffer. **Nothing is indexed for long-term retrieval yet.**

Step B: The Crystal Compressor (Topological Distillation)

Triggered asynchronously (analogous to sleep/rest states), the Compressor LLM processes the buffer. It is prompted to act as a topological filter:

1. Identify the logical skeleton (the invariant).
2. Discard the specific variables, numbers, and domain context (the deformation).
3. Output a strictly formalized, 2-to-3 sentence generalized axiom or algorithmic trick.

Step C: Semantic Manifold Storage (One-Form Optimization)

When storing the extracted axiom, we don't just append it. We check if it is a slight variation of an existing axiom. We define an energy functional representing the redundancy in the database. Using your differential calculus approach, we calculate the exterior derivative (a one-form) of this energy space. If the gradient points toward an existing node, the axioms are merged, continually tightening the AI's understanding rather than just adding to a list.

Step D: Context-Injection (Retrieval)

When a new problem arrives (e.g., your 10-word Lie Group prompt), the system queries the Semantic Manifold. Because the manifold only contains compressed, topological invariants, the retrieval returns something like:

"Axiom: When calculating invariants on a curved space, use the Laplace-Beltrami operator to generalize the divergence of the gradient."

This 20-word densely packed axiom is injected into the Master Agent's context. The LLM's attention mechanism remains entirely focused on your 10-word prompt, perfectly augmented by the 20-word hint, completely avoiding context-collapse.